

Basic Instructions and How To Privacy Analysis of Augmented Reality Glasses

1 WiFi Traffic

1.1 Basic Set Up



1. Plug in the ALFA network adapter with a USB adapter. The network adapter used in this experiment is the Long-Range USB Adapter of model AWUS036NHA.
2. Make sure the laptop is connected to an internet connection.
3. Start the hotspot (I used a bash script provided to me).
3. Connect the phone to the hotspot.
4. Start wireshark with sudo access, if it is not installed:

```
sudo wireshark
```

If wireshark is not installed, write:

```
sudo apt install wireshark
```

5. Choose the interface of the network adapter and start capture.
6. Start experiments by interact with the glasses and the phone to generate traffic.

2 Bluetooth Traffic

2.1 Bluetooth sniffer

Three different Bluetooth Sniffer devices were considered during this project:

- BBC Micro:bit V2.20/21
- Adafruit nRF51822 Bluefruit
- Ubertooth one

2.2 Basic Set Up

The installation for each device can be found in the Bluetooth Sniffer file in the current folder.

3 How to Live Stream on Instagram

1. Power on the glasses and make sure they are connected on the Meta View app.
2. Open Instagram.
3. Press the  at the bottom middle.
4. Then swipe left gently till the end where you see 'Live' at the bottom.
5. Make sure to check who do you want seeing the live stream. (Default is everyone)
6. On the right side of the phone, you should see your glasses, press that.
7. Then double press the capture button on your glasses to switch to your glasses camera. Meta should also inform you about this step.
8. Start live stream at the bottom middle circle button.

Note that you can switch between cameras at any time by double pressing the capture button on your glasses.

4 Unpairing the glasses

Either: In Settings,

1. Go to Bluetooth and unpair the glasses;

Or: In the Meta View app,

1. Press settings at the bottom right.
2. Tap 'Your glasses' or directly the device you want to unpair.
3. Tap 'General', 'Unpair glasses', 'Unpair'.

Note that the first option is more preferred and easier.

5 Re-pairing the glasses to the phone

<https://www.meta.com/en-gb/help/smart-glasses/articles/ray-ban-meta/set-up-ray-ban-meta-smart-glasses/>

1. Open the Meta View app.
2. Go to the settings at the bottom right and press 'Add Device' on the top corner of the app.
3. Choose the Meta Ray Ban glasses.
4. Take the glasses out of the case and power on.
5. Put the glasses back inside the case and close the case.
6. Wait a bit for the LED on the case to start blinking blue, this means that the device is ready to pair with the Meta View app. If the LED does not blink blue, press and hold the button on the back of the case for at least 5 seconds till you see blinking blue.
7. On the Meta View app, press 'Blue LED light turned on' then follow the on-screen instructions.

6 Factory Reset the glasses

Link: <https://www.meta.com/en-gb/help/smart-glasses/articles/ray-ban-meta/factory-reset-ray-ban-meta-smart-glasses/>

1. Power on the glasses and only need to close the right arm.
2. Press the capture button 5 times very quickly then the notification LED should turn orange.
3. Press and hold the capture button until that notification LED starts blinking.
4. Then release the capture button wait for the LED to turn blue that indicates that you can now pair to a new device.